

Foundation Models for Precision Classification in NSBI: toward a Higgs self-coupling (κ_λ) measurement in $HH \rightarrow b\bar{b}\tau\tau$

working note — draft for discussion

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Bottom line

For constraining κ_λ , the statistically-limited piece is *kinematic shape* discrimination of nearly-identical distributions. Both our own tests and the literature indicate this piece is **method-independent and statistics-limited**: a foundation model (FM) will *not* out-classify a good high-level-feature NSBI on the κ_λ shape (*though a systematics-robust FM can still help the κ_λ inference under profiling — see §6*). The genuine leverage of an FM is elsewhere — (i) **data efficiency** at low signal stats, (ii) **signal-vs-background acceptance/purity**, and (iii) **robustness to systematics**. The recommended path is a *factorized, calibration-first* strategy: keep the validated signal-extraction machinery and do NSBI on high-level kinematics now; pursue a jet-free / event-level FM on the acceptance axis as a staged, calibration-gated upgrade.

1 Physics goal and current status

We aim to constrain the Higgs trilinear self-coupling modifier $\kappa_\lambda = \lambda_{HHH}/\lambda_{HHH}^{\text{SM}}$ through non-resonant di-Higgs production in the $HH \rightarrow b\bar{b}\tau\tau$ final state, with the dominant sensitivity in the **resolved regime** ($m_{HH} \in [250, 400]$ GeV). κ_λ enters as a shape distortion of m_{HH} (interference between the triangle, $\propto \lambda_{HHH}$, and box amplitudes in ggF).

A neutral-simulation-based-inference (NSBI) fit using **10 event-level features**,

$$m_{HH}, \cos\theta^*, p_T^{HH}, m_{bb}, \Delta R_{bb}, m_{\tau\tau}, \Delta\phi_{HH}, \Delta R_{\tau\tau}, p_T^{H_1}, p_T^{H_2},$$

already **tightens the κ_λ 95% CI by roughly a factor of two** (with $t\bar{t}$ background only, so far). The central *challenge* is statistical: very low signal yield, made worse once systematic uncertainties are profiled.

Hypothesis under test: a foundation-model + fine-tuning paradigm for the density-ratio (precision-classification) tasks inside the NSBI could improve the constraint — primarily via data efficiency and better object use.

2 The key decomposition: a κ_λ NSBI is two ML jobs

The single most clarifying point is that the measurement contains two distinct learning problems, which an FM helps very differently:

The κ_λ *constraint* comes from job (B). Job (A) sets the effective signal purity and is where the analysis is acceptance- and systematics-limited. A flavor-pretrained FM is built for (A), not (B) — which explains the results below.

Job	Discriminates	Information needed / best tool
(A) Signal vs. background	$HH \rightarrow b\bar{b}\tau\tau$ vs. $t\bar{t}$ /QCD/fakes	<i>Object identity</i> : b-tagging, real-vs-fake τ , substructure — flavor FM (Sophon-ak4) territory .
(B) κ_λ shape / inference	$\kappa_\lambda = 0$ vs. 5 vs. ... (morphing)	<i>Event kinematics</i> (m_{HH} shape) — high-level features / NSBI; a “precision classification of nearly-identical distributions” problem.

3 What we tested, and what we found

We adapted the `nsbi-lhc-toolkit` with a vendored Particle Transformer (ParT) backbone and **Sophon-ak4** (the AK4/resolved-jet ParT foundation model, pretrained for jet-flavor tagging on JetClass-II). A hierarchical model encodes each jet’s constituents and aggregates jets+objects with an event set-transformer. Three controls were compared on the $\kappa_\lambda=0$ **vs.** $\kappa_\lambda=5$ task (with $\kappa_\lambda=5$ as the fixed reference/denominator): `scratch` (random-init ParT), `sophon_frozen`, `sophon_finetune`.

Table 1: *
Validation weighted-BCE (lower = better; random floor = $\ln 2 = 0.693$).

N / point	scratch	sophon_finetune	sophon_frozen
100k (40 ep) [†]	0.6378	0.6351	0.6486
2,000	0.6752	0.6775	0.6798
5,000	0.6387	0.6457	0.6538

[†]earlier configuration (before early-stopping and the lowered fine-tune LR); single seed at low N (error bars pending).

Empirical finding

On the κ_λ *shape* task, pretraining does **not** help — the ordering is consistently **scratch** $\leq$ **finetune** $<$ **frozen**. Frozen flavor features are the worst; fine-tuning merely relearns toward scratch.

Interpretation. (i) *Pretext mismatch*: Sophon-ak4 is optimized for jet flavor, whose per-jet embeddings are lossy about the p_T /mass needed to reconstruct m_{HH} from cross-jet correlations. (ii) The task is precisely “precision classification of nearly-identical distributions,” which (next section) is statistics-limited and largely architecture-independent.

4 External evidence and the FM landscape

A 2025 review [1] organizes HEP foundation models into *supervised* (OmniLearn, Sophon, RS3L) and *self-supervised* (OmniJet- α , MPM) families. Points most relevant here:

- **Explicit vs. implicit priors are a wash — and statistics-limited** [2]. A direct comparison of L-GATr (Lorentz-equivariant) vs. OmniLearn (FM pretrained on 100M events) on *precision classification of similar classes* finds “comparable performance ... given the statistical precision of the datasets,” and that the gains from encoding physics are “method-independent.” This benchmark *is* our κ_λ -shape task, and predicts our null result. Nuances:

on the $H1/ep$ task the FM *beat* the equivariant net, and for small signal injections the implicit (FM) approach was slightly better — a mild lean toward FMs in the low-statistics corner.

- **Taus are not a blocker.** OmniJet- α fine-tuned for hadronic- τ reconstruction improved momentum resolution by $\sim 50\%$ vs. from-scratch training [3] — so an earlier worry that OmniLearn-style FMs are unsuited to τ final states was premature.
- **Event-level FMs exist.** PECM [4] is a GNN over event objects (jets, e , μ , γ , MET with four-momenta + type IDs) pretrained on 120M events; it reports >5 percentage-point accuracy gains at 10^3 – 10^5 events/class (our regime) and $\sim 12\times$ faster fine-tuning, transferring across simulators. Closest match to our event-level inputs.
- **Self-supervision retains kinematics; RS3L targets systematics.** MPM [5] learns task-agnostic representations (less biased than supervised flavor labels). RS3L [6] pretrains on *re-simulation* augmentations (varied shower/hadronization/detector), yielding embeddings *robust to those systematics* — directly relevant to a systematics-limited measurement.

5 Two architectural strategies for the measurement

Strategy A — jet-free, end-to-end. Feed *all* particle-flow candidates of the event (no jet clustering) into a ParT/event-FM, add the 10 high-level features, and do NSBI for κ_λ in one model. Motivated by a recent *jet-free* $HH \rightarrow 4b$ analysis [7]: ingesting all PF candidates (~ 300 /event) into a ParT gave $>5\times$ search significance and an order-of-magnitude more retained signal at high background rejection. Caveats: it is *not* an FM (task-specific, trained on $\sim 10^8$ events — “training statistics play a critical role”), it is $HH \rightarrow 4b$ (a worse case for jet methods than $b\bar{b}\tau\tau$), and it required a *new* “calibrate–validate–measure” scheme (calibrate on $ZZ \rightarrow 4b$, validate on $ZH \rightarrow 4b$) because object-level scale factors do not apply to a jet-free discriminant.

Strategy B — factorized. Use the current CMS-style signal extractor for job (A), then NSBI on the high-level features (or an FM embedding) for job (B). The CMS $HH \rightarrow b\bar{b}\tau\tau$ analysis [8] already uses a sophisticated DNN: a Lorentz Boost Network front-end, a $\tau\tau$ neutrino-regression subnet, and a DenseNet classifier, with inputs that are full-system four-momenta *plus* ParticleNet/U-ParT (b-tag, c-vs-b, c-vs-l) and HHbTag scores — i.e. it already uses ParT as fixed tagger *scores* on reconstructed objects, with the entire validated systematics machinery built around those objects.

	A: jet-free FM (+10 feats) $\rightarrow$ NSBI	B: CMS S/B $\rightarrow$ NSBI on feats/FM
Information ceiling	Highest (soft radiation, no clustering/WP losses)	High, capped by reconstruction
Where it wins	Signal acceptance & purity	Clean κ_λ -kinematics (the proven $\sim 2\times$)
κ_λ -shape info	Same (kinematic)	Same (kinematic)
Calibration / systematics	Hard & novel (new scheme)	Validated CMS machinery
NSBI tractability	High-dim, harder closure/morphing	Low-dim, interpretable, easy closure
Data efficiency	Needs huge stats unless pretrained	Already efficient
Risk / time-to-result	High / frontier	Low / near-term

Important caveat: “Sophon-ak4 jet-free” mixes scales — Sophon-ak4 was pretrained on *single AK4 jets* (~ 128 constituents), so applying it to the *whole event* (~ 300 particles) is out-of-distribution. Jet-free-whole-event calls for an *event-scale* FM (PECM-style, or a from-scratch / SSL event-ParT), not Sophon-ak4 stretched beyond its domain.

6 Where the FM actually helps: separation vs. systematics robustness

A natural but *over-simplified* reading of the two-job decomposition is “the FM belongs in the s/b ratio, not the κ_λ ratio.” That holds for one benefit axis and fails for the other — and the one it fails for is precisely the systematics-limited regime that constrains this measurement. An NSBI κ_λ fit needs (at least) two learned ratios: a signal-vs-background ratio $s(x)/b(x)$ and a κ_λ -parametrized signal-shape ratio $r(x; \kappa_\lambda)$. An FM is a *backbone for a ratio estimator*, so the question is always “which ratio, and helping *what*?”

- **Axis 1 — separation / central sensitivity.** How well a ratio discriminates (the kl0-vs-kl5 ceiling above). Here the 10 features are at ceiling and data-efficient, so an FM backbone does *not* improve the κ_λ ratio’s discrimination; its separation gain accrues only to s/b (acceptance, object ID).
- **Axis 2 — robustness + estimator stability under limited stats.** A property of the learned *representation*, not of its separation power. It can improve *either* ratio — including κ_λ — and it is the axis that matters when, as here, the measurement is systematics- and statistics-limited.

	Axis 1: separation	Axis 2: robustness / stats stability
s/b ratio	FM helps (acceptance, object ID)	FM helps (robust + data-efficient)
κ_λ shape ratio	FM does <i>not</i> help (at ceiling)	FM <i>can</i> help (robust rep $\rightarrow$ smaller shape syst.)

How an FM helps the κ_λ ratio on Axis 2:

1. *Robustness* (e.g. RS3L [6]): a representation invariant to shower/hadronization/detector variations yields a discriminant whose *shape* shifts less under nuisances $\rightarrow$ smaller profiled systematics $\rightarrow$ tighter CI *even at fixed separation*.
2. *Data-efficient systematic templates*: building systematic up/down variations needs ratios estimated on stats-starved varied MC; a pretrained, data-efficient estimator gives less noisy templates $\rightarrow$ better-behaved profiling. (Largest where the estimator is high-dimensional; smaller for the already-efficient 10-feature ratio.)
3. (*Advanced*) *nuisance-parametrized NSBI*: learn $r(x; \kappa_\lambda, \theta_{\text{syst}})$ directly; an FM backbone makes the higher-dimensional conditional learning more data-efficient.

Catches (not free):

- To get the Axis-2 benefit on κ_λ you build the ratio on the FM representation rather than the at-ceiling hand features — trading interpretability / easy closure for robustness.
- You swap *well-calibrated object* systematics (JES, τ energy scale) for the FM representation’s *own* systematics, which must then be established (the calibrate–validate–measure burden); robustness to known systematics can introduce new ones.
- Whether it is a *net* win is empirical — and it **will not appear in BCE/AUC**, which measure separation. We have so far measured only separation, which by construction cannot reveal a robustness gain.

Corrected conclusion

The FM’s *separation* benefit accrues only to s/b ; its *robustness* / *data-efficiency* benefit can accrue to *either* ratio, including κ_λ . So an FM will not beat the high-level features on κ_λ *separation*, but a systematics-robust / data-efficient FM (RS3L-style) can still help the κ_λ *measurement* by making the inference more robust under profiling when stats are limited — the one place an FM earns its keep on the kinematic side.

The diagnostic this implies. The right test of the FM’s value here is *not* classifier loss but *robustness*: propagate the dominant nuisances (JES, τ energy scale, parton shower) through each control and measure (i) how much the discriminant distribution shifts (nominal vs. varied) and (ii) whether the *profiled* κ_λ CI shrinks. A feature-based vs. FM-based comparison on *those* metrics — not BCE/AUC — is what would show an FM helping where the analysis is actually limited.

7 Synthesis and recommendation

Key insights

1. The κ_λ constraint is **kinematic and statistics-limited**; explicit (equivariant) and implicit (FM) priors give comparable shape performance. Do not expect an FM to beat the high-level-feature NSBI on the κ_λ shape.
2. The FM’s real value is **data efficiency**, **S/B acceptance/purity**, and **systematics robustness** — not shape extraction.
3. For a systematics-limited result, **calibration of any learned discriminant is the dominant risk**; the jet-free upside can be eaten by a poorly understood systematic.
4. Use the right FM for the right job: Sophon-ak4 (jet-scale) $\rightarrow$ per-jet S/B; event-scale FM $\rightarrow$ jet-free acceptance; high-level features / NSBI $\rightarrow \kappa_\lambda$.

Recommended staged plan.

- **Near-term (low risk):** Strategy B, sharpened. Do NSBI on the high-level features (optionally + the CMS DNN score) within the validated signal regions — the direct productization of the $\sim 2\times$ result on real, calibrated data.
- **Test the right task:** a signal-vs- $t\bar{t}$ diagnostic (3 controls) — where per-jet Sophon *should* help, unlike the κ_λ -shape task. Plus a high-level kinematic ceiling baseline (BDT) to bound how much shape headroom exists.
- **Parallel R&D (high ceiling):** a jet-free / *event-level* FM for acceptance, introduced as a *calibratable* added discriminant (not a replacement), folded into the NSBI only once its calibration is validated.
- **Distinctive contribution:** RS3L-style re-simulation pretraining for a *systematics-robust* embedding — aimed squarely at the systematics bottleneck, and evaluated via the robustness diagnostic of §6 (nuisance shape variation + profiled CI), not BCE/AUC.

8 Open questions

- Do explicit and implicit priors learn the *same* physics, and does combining them (a Lorentz-equivariant *foundation* model) help in our regime?
- How large is the jet-free acceptance gain for $b\bar{b}\tau\tau$ (cleaner than $4b$), and can it be calibrated with $Z \rightarrow \tau\tau/Z \rightarrow b\bar{b}$ control samples?
- Does any S/B or acceptance gain actually translate into a tighter κ_λ CI once systematics are profiled (the figure of merit is the constraint, not BCE/AUC)?

References

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